

先进计算与数字工程研究中心



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reaches the sorting gate

## The observation

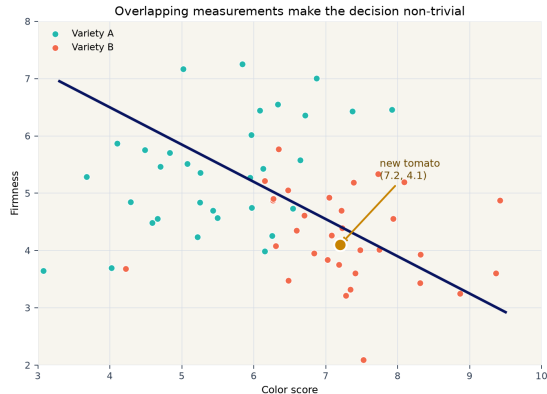
Color score  $x_1 = 7.2$  and firmness  $x_2 = 4.1$ .

## The decision

Pack now or hold for another day?

## The complication

Two varieties overlap in the sensor space.



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| Question                             | Quantity              | Typical use         |
|--------------------------------------|-----------------------|---------------------|
| How could this measurement arise?    | $p(x   y)$ and $p(y)$ | Describe each class |
| Which label fits this measurement?   | $p(y   x)$            | Predict a class     |
| Which side of the boundary is it on? | $g(x)$                | Make a decision     |



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| Syllabus topic                | Question answered                          |
|-------------------------------|--------------------------------------------|
| Classifier families           | What should the model represent?           |
| Linear discriminant functions | What does one straight boundary mean?      |
| Generalized linear functions  | How can a feature map create a curve?      |
| Piecewise linear functions    | How can several local lines form one rule? |
| Fisher linear discriminant    | Which projection separates the classes?    |

NumPy examples will calculate scores, maps, prototypes, and Fisher directions.

# World model and decision model

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## Generative route

Fit  $p(x | y)$  and  $p(y)$ , then derive  $p(y | x)$ .

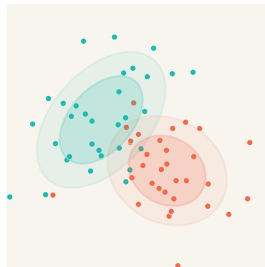
## Discriminative route

Fit  $p(y | x)$  or a score  $g(x)$  directly.

## Do not rank them by name

The useful choice depends on data, assumptions, and the required output.

Generative: describe each class



$p(x | y)$  and  $p(y)$

Discriminative: describe the decision



$p(y | x)$  or  $g(x)$

Fit a color and firmness distribution for each maturity class.

$$p(y = k | x) = \frac{p(x | y = k)p(y = k)}{\sum_j p(x | y = j)p(y = j)}$$

**What the model can do.** Classify, sample plausible measurements, and reveal unusual observations.

# Discriminative modelling targets the label

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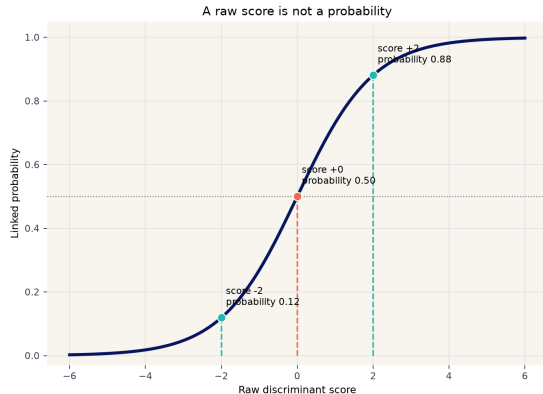
## Conditional probability

Estimate  $p(y | x)$  when calibrated uncertainty matters.

## Decision score

Estimate  $g(x)$  when only the class decision is required.

$$\hat{y} = \arg \max_k p(y = k | x) \quad \text{or} \quad \hat{y} = \text{sign } g(x)$$



# Bayes can produce a discriminant function

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## Start from the posterior

Choose class 1 when  $p(y = 1 | x) > p(y = 2 | x)$ .

$$g(x) = \log p(x | y = 1) + \log p(y = 1) - \log p(x | y = 2) - \log p(y = 2)$$



## Same decision

Class 1 is selected when  $g(x) > 0$ .

# What unlabeled inputs can contribute

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## Possible generative use

Unlabeled  $x$  values can help estimate the structure of  $p(x)$  under a stated model.

## No automatic benefit

A wrong mixture or cluster assumption can move the boundary in the wrong direction.

## Purely supervised discriminative fit

Without an added objective, unlabeled rows contribute no label loss.

# Fewer assumptions can focus capacity on the boundary

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## A direct target

A discriminative model need not reproduce every detail of  $p(x)$ .

## Tomato sensor

Lighting variation that does not change the label may be ignored by the boundary.

## The tradeoff

Less structure also means fewer ways to use missing labels or detect implausible inputs.

# When generative assumptions help

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| Situation                            | Why the structure helps                                |
|--------------------------------------|--------------------------------------------------------|
| Few labels and credible class models | Parameters share information through the model         |
| Missing measurements                 | The joint distribution can marginalize missing entries |
| Anomaly screening                    | Low likelihood can flag an unfamiliar input            |
| Simulation is required               | The fitted model can produce synthetic observations    |

# When generative assumptions hurt

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## Misspecified shape

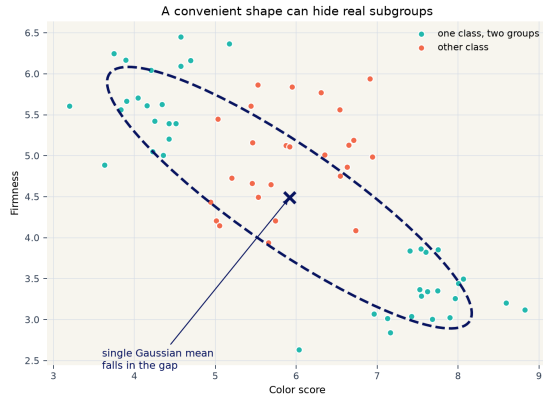
A Gaussian ellipse cannot represent every class distribution.

## Hidden groups

Two tomato varieties may create two modes inside the same maturity class.

## Consequence

The estimated likelihood can be wrong far from the boundary and still alter the decision.



# A fair comparison uses the same evidence

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- 1 Use the same training, validation, and test split.
- 2 Tune both model families on the validation set.
- 3 Evaluate the metric that matches the decision cost.
- 4 Inspect calibration only when probabilities are claimed.



## Common mistake

Comparing one tuned model with another model's default settings proves little.

# Case change under a different light

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## Morning

Color and firmness separate the two decisions with a simple diagonal boundary.

## Afternoon

A warm lamp shifts the color score by  $+1.5$  for both classes.

## Question

Which part should change: the data model, the features, or the boundary bias?

## family choice

| Need                                                 | First model to consider       |
|------------------------------------------------------|-------------------------------|
| A direct boundary with many labels                   | Discriminative                |
| A class distribution and simulated inputs            | Generative                    |
| Useful unlabeled structure with credible assumptions | Generative or semi supervised |
| Only calibrated class probabilities                  | Conditional discriminative    |



Choose one row and state the assumption that makes your choice reasonable.

# Four questions for every boundary

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- 1 What representation enters  $g(x)$ ?
- 2 What shape can  $g(x) = 0$  express?
- 3 How does the score become a class decision?
- 4 What evidence would reveal a bad assumption?



Use this checklist

Apply the same four questions to linear, mapped, piecewise, and Fisher boundaries.

## discriminant functions

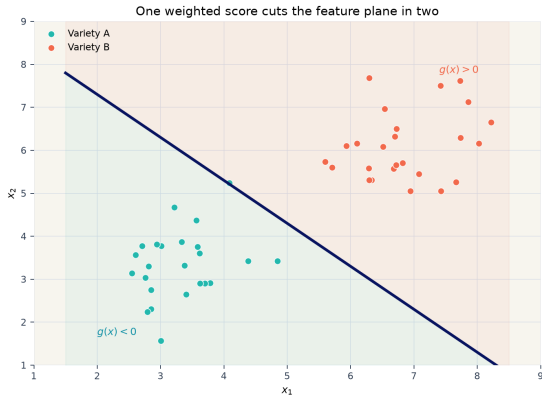
## The model

A weighted sum turns a feature vector into one score.

$$g(x) = w^\top x + w_0$$

## The boundary

All points satisfying  $g(x) = 0$ .





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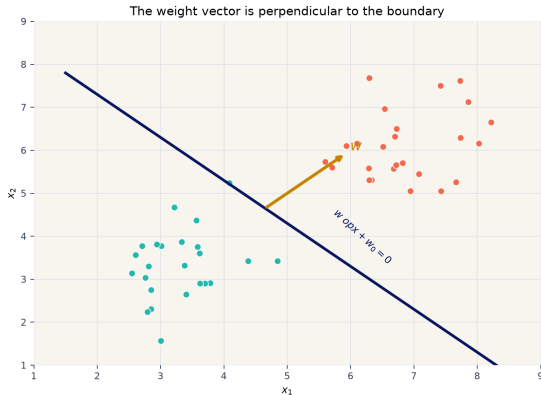
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100%

For boundary points  $x_a$  and  $x_b$ ,  $w^\top (x_a - x_b) = 0$ .

$$w \perp (x_a - x_b)$$

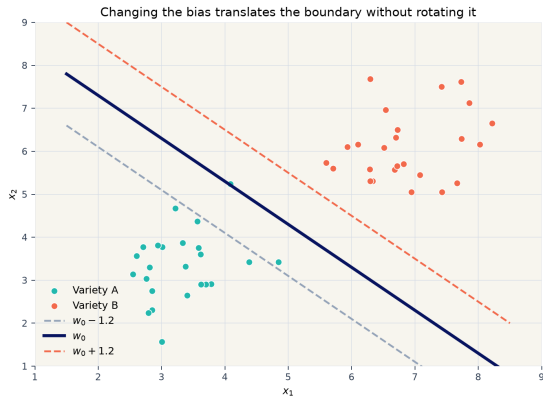
Moving in the direction  $w$  increases  $g(x)$  fastest.



Changing  $w_0$  moves the boundary parallel to itself.

$$\text{shift along } w = -\frac{\Delta w_0}{\|w\|}$$

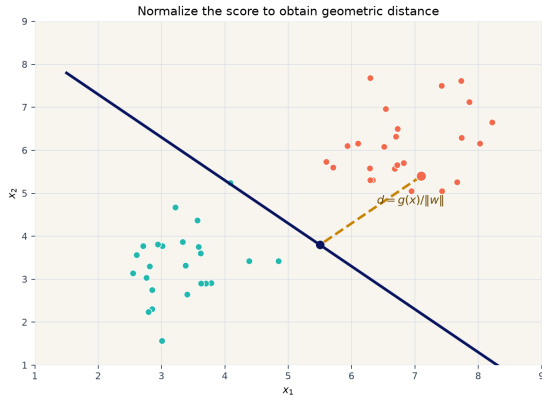
A constant color offset can often be absorbed by the bias.



$g(x) = w^\top x + w_0$  changes if every parameter is rescaled.

$$d(x) = \frac{w^T x + w_0}{\|w\|}$$

$g(x)$  and  $10g(x)$  define the same boundary but not the same raw magnitude.



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$$w = (1, -0.8)^\top, w_0 = -3.5, x = (7.2, 4.1)^\top$$

$$g(x) = 7.2 - 0.8 \times 4.1 - 3.5 = 0.42$$

The score is positive, so the model selects pack now.

**Distance.**  $0.42/\sqrt{1^2 + 0.8^2} \approx 0.328$  feature units.

# Feature units change the weights

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## Two equivalent records

Firmness 4.1 on a 0 to 10 scale equals 41 on a 0 to 100 scale.

## Equivalent model

The weight  $-0.8$  becomes  $-0.08$  when that feature is multiplied by 10.

## Wrong shortcut

Keeping the old weight after changing units moves the boundary.

# Standardization makes scales comparable

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## Transform each feature

$$z_j = \frac{x_j - \mu_j}{s_j}$$

## Interpretation

A one unit change in  $z_j$  means one training standard deviation.

## Deployment rule

Reuse the training values of  $\mu_j$  and  $s_j$  for new samples.

# The augmented vector absorbs the bias

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## Augmented feature

$$\tilde{x} = (x_1, \dots, x_D, 1)^\top$$

## Augmented weight

$$\tilde{w} = (w_1, \dots, w_D, w_0)^\top$$

$$g(x) = \tilde{w}^\top \tilde{x}$$

## Why it helps

One inner product handles both feature weights and the bias.

# Equivalent parameters and one boundary

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## Positive rescaling

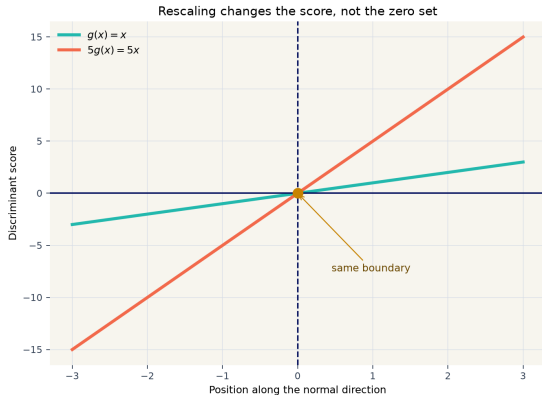
$(w, w_0)$  and  $(cw, cw_0)$  define the same decision for every  $c > 0$ .

## Negative rescaling

For  $c < 0$ , the boundary stays but the class labels swap.

## Identifiability

A normalization or training objective selects one representative from many equivalent parameter sets.



Separable training points do not guarantee perfect predictions on future data.

# Exclusive or cannot use one straight boundary

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| $x_1$ | $x_2$ | Positive class |
|-------|-------|----------------|
| 0     | 0     | Yes            |
| 0     | 1     | No             |
| 1     | 0     | No             |
| 1     | 1     | Yes            |



## Contradiction

No affine function can place both diagonal corners on one side and the other corners on the opposite side.

## Later solutions

Add an interaction feature, combine local lines, or use a deeper model.



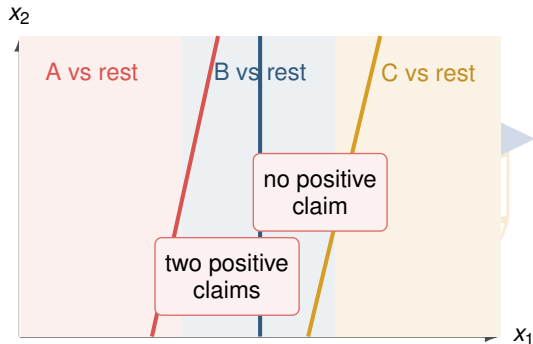
Several classes claim the same sample.

No positive score

No classifier claims the sample.

## Required policy

Use comparable scores, a priority rule, or a reject option. State the choice explicitly.



Independent binary rules need a policy for overlaps and gaps.



One versus one classification

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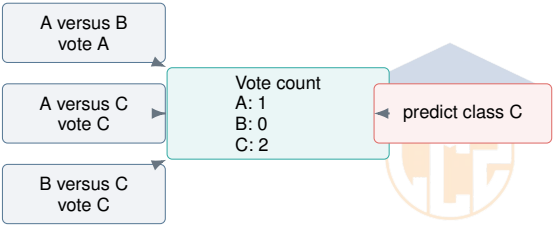
Pairwise rule

Fit one classifier for every pair of classes.

number of classifiers =  $\frac{C(C - 1)}{2}$

Prediction

Each pair casts one vote. The class with most votes wins.



$C(C - 1)/2$  pairwise classifiers; ties need a declared rule.

# Pairwise voting can tie

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## Three pairwise results

A beats B, B beats C, and C beats A.

| Class | Votes |
|-------|-------|
| A     | 1     |
| B     | 1     |
| C     | 1     |



## A complete classifier needs a tie rule

Examples include summed margins, calibrated pairwise probabilities, or a fixed class order.

# One discriminant score per class

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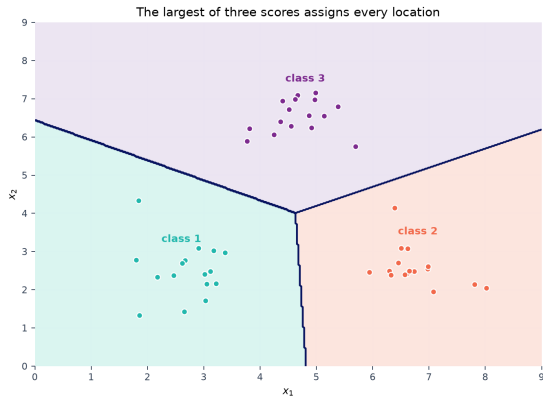
## Shared comparison

Fit  $C$  scores and choose the largest.

$$\hat{y} = \arg \max_c g_c(x), \quad g_c(x) = w_c^\top x + w_{c0}$$

## Pairwise boundary

Classes  $c$  and  $k$  meet where  $g_c(x) = g_k(x)$ .



# Argmax partitions the whole plane

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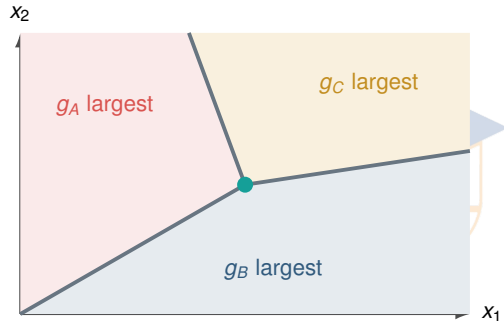
## Complete assignment

Every point has at least one largest score.

## Ties

Two or more classes can share the maximum exactly on a boundary.

**Geometry.** The visible class regions are intersections of linear inequalities.



Every point receives the class with the largest score; ties lie on boundaries.

# A numeric three class decision

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## Scores

$$g_1 = -x_1 + x_2, g_2 = x_1 + x_2 - 5, g_3 = -x_2 + 1$$

| Input  | $g_1$ | $g_2$ | $g_3$ | Argmax  |
|--------|-------|-------|-------|---------|
| (6, 5) | -1    | 6     | -4    | Class 2 |
| (3, 5) | 2     | 3     | -4    | Class 2 |



## One versus rest difference

At (3, 5), both  $g_1$  and  $g_2$  are positive.

# Classifier count grows differently

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| Classes | One versus rest | One versus one | Comparable class scores |
|---------|-----------------|----------------|-------------------------|
| 3       | 3               | 3              | 3                       |
| 10      | 10              | 45             | 10                      |
| 100     | 100             | 4,950          | 100                     |



## Cost beyond training

Prediction must also evaluate the required scores or pairwise classifiers.

# Scores and probabilities answer different questions

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| Output                 | What it supports               | What it does not guarantee     |
|------------------------|--------------------------------|--------------------------------|
| Score $g(x)$           | Ranking and boundary decisions | Calibrated probability         |
| Margin distance        | Geometric separation           | Frequency among similar cases  |
| Probability $p(y   x)$ | Expected risk and thresholds   | Correctness without validation |

# A link function can convert a score

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## Logistic example

$$p(y = 1 \mid x) = \frac{1}{1 + \exp[-g(x)]}$$

## Boundary

$g(x) = 0$  corresponds to probability 0.5.

## Preview only

The link defines a probability model. Its fitting and evaluation appear in the linear models lecture.

# Least squares can fit a discriminant

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## Targets

Encode classes as  $t_i \in \{-1, +1\}$ .

## Objective

$$\min_{\tilde{w}} \|\tilde{X}\tilde{w} - t\|_2^2$$

## Closed form when appropriate

Solve  $\tilde{X}^\top \tilde{X}\tilde{w} = \tilde{X}^\top t$ .

# Least squares uses every residual

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## Correct but extreme

A positive sample with target  $+1$  and score  $+8$  has residual  $7$ .

## Wrong but near

A positive sample with score  $-0.1$  has residual  $-1.1$ .

## Mismatch

The first point is correctly classified but contributes a much larger squared error.

# A large residual can pull the boundary

## Before the extreme point

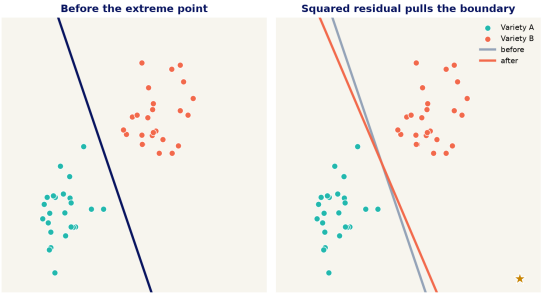
The fitted line follows the main cloud.

## After the extreme point

Squared loss gives the distant residual disproportionate influence.

## Response

Inspect leverage, measurement errors, and a classification specific loss.



# NumPy batch scoring

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```
import numpy as np

X = np.array([[7.2, 4.1],
              [5.4, 6.0],
              [6.8, 3.5]])
w = np.array([1.0, -0.8])
b = -3.5

scores = X @ w + b
labels = np.where(scores > 0, 1, -1)
print(scores)
print(labels)
```

Expected scores

[0.42, -2.90, 0.50] and labels [1, -1, 1].

# NumPy multiclass scores

```
W = np.array([[ -1.,  1.],
               [ 1.,  1.],
               [ 0., -1.]])
b = np.array([0., -5., 1.])
X = np.array([[6., 5.], [3., 5.]])

scores = X @ W.T + b
prediction = scores.argmax(axis=1) + 1
print(scores)
print(prediction)
```

## Axis check

Rows are samples and columns are class scores. Take the maximum across columns.

Transform  $x$  into features  $\phi(x)$ , then use a linear score in the transformed space.

$$g(x) = w^\top \phi(x)$$

A hyperplane in feature space can appear curved in the original space.

$$g(x) = w_0 + w_1\phi_1(x) + \cdots + w_M\phi_M(x)$$

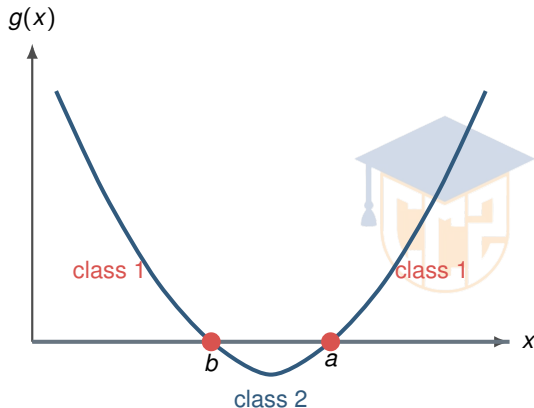
If  $\phi_1(x) = x^2$ , the score is nonlinear in  $x$ .

Generalized linear here means a fixed feature map followed by a linear discriminant. It is distinct from the statistical family called generalized linear models.

Class 2 occupies  $b < x < a$ . Class 1 occupies  $x < b$  or  $x > a$ .

A one dimensional affine score changes sign at most once.

The desired rule needs two crossing points.



Two zero crossings isolate the middle interval.

# Lift the input into two features

## Feature map

$$\phi(x) = (x^2, x, 1)^\top$$

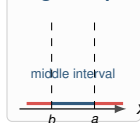
$$g(x) = (x - a)(x - b) = x^2 - (a + b)x + ab$$

## Linear weights

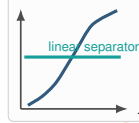
$$w = (1, -(a + b), ab)^\top$$

**Decision.** Positive outside the interval and negative inside when  $a > b$ .

### Original input



### Lifted features



The quadratic map turns two crossings on the line into one linear separator in feature space.

Let  $z_1 = x^2$  and  $z_2 = x$ .

$$z_1 - (a + b)z_2 + ab = 0$$

This is a line in the  $z$  plane and a quadratic equation on the original  $x$  axis.

# Exclusive or becomes linear with one interaction

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| $x_1$ | $x_2$ | $x_1 x_2$ | Target $y$ |
|-------|-------|-----------|------------|
| -1    | -1    | +1        | +1         |
| -1    | +1    | -1        | -1         |
| +1    | -1    | -1        | -1         |
| +1    | +1    | +1        | +1         |



## A one feature rule

$g(x) = x_1 x_2$  and  $\hat{y} = \text{sign } g(x)$ .

## What changed

The interaction records whether the two signs agree.

$$\phi(x) = (x_1^2, x_1x_2, x_2^2, x_1, x_2, 1)^\top$$

$$g(x) = w_{11}x_1^2 + w_{12}x_1x_2 + w_{22}x_2^2 + w_1x_1 + w_2x_2 + w_0$$

Ellipses, parabolas, hyperbolas, pairs of lines, and degenerate cases.

# Quadratic feature count

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| Component       | Number             |
|-----------------|--------------------|
| Quadratic terms | $D(D + 1)/2$       |
| Linear terms    | $D$                |
| Bias            | 1                  |
| Total           | $(D + 1)(D + 2)/2$ |



For two inputs

$3 + 2 + 1 = 6$  features.

For one hundred inputs

5,151 features.

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All monomials up to degree  $r$

$$M = \begin{pmatrix} D + r \\ r \end{pmatrix}$$

| Input dimension $D$ | Degree $r$ | Features $M$ |
|---------------------|------------|--------------|
| 10                  | 2          | 66           |
| 10                  | 3          | 286          |
| 100                 | 2          | 5,151        |
| 100                 | 3          | 176,851      |

## Capacity warning

Higher degree can overwhelm both memory and the available sample size.

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1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2030, 2031, 2032, 2033, 2034, 2035, 2036, 2037, 2038, 2039, 2040, 2041, 2042, 2043, 2044, 2045, 2046, 2047, 2048, 2049, 2050, 2051, 2052, 2053, 2054, 2055, 2056, 2057, 2058, 2059, 2060, 2061, 2062, 2063, 2064, 2065, 2066, 2067, 2068, 2069, 2070, 2071, 2072, 2073, 2074, 2075, 2076, 2077, 2078, 2079, 2080, 2081, 2082, 2083, 2084, 2085, 2086, 2087, 2088, 2089, 2090, 2091, 2092, 2093, 2094, 2095, 2096, 2097, 2098, 2099, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2113, 2114, 2115, 2116, 2117, 2118, 2119, 2120, 2121, 2122, 2123, 2124, 2125, 2126, 2127, 2128, 2129, 2130, 2131, 2132, 2133, 2134, 2135, 2136, 2137, 2138, 2139, 2140, 2141, 2142, 2143, 2144, 2145, 2146, 2147, 2148, 2149, 2150, 2151, 2152, 2153, 2154, 2155, 2156, 2157, 2158, 2159, 2160, 2161, 2162, 2163, 2164, 2165, 2166, 2167, 2168, 2169, 2170, 2171, 2172, 2173, 2174, 2175, 2176, 2177, 2178, 2179, 2180, 2181, 2182, 2183, 2184, 2185, 2186, 2187, 2188, 2189, 2190, 2191, 2192, 2193, 2194, 2195, 2196, 2197, 2198, 2199, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2209, 2210, 2211, 2212, 2213, 2214, 2215, 2216, 2217, 2218, 2219, 2220, 2221, 2222, 2223, 2224, 2225, 2226, 2227, 2228, 2229, 2230, 2231, 2232, 2233, 2234, 2235, 2236, 2237, 2238, 2239, 2240, 2241, 2242, 2243, 2244, 2245, 2246, 2247, 2248, 2249, 2250, 2251, 2252, 2253, 2254, 2255, 2256, 2257, 2258, 2259, 2260, 2261, 2262, 2263, 2264, 2265, 2266, 2267, 2268, 2269, 2270, 2271, 2272, 2273, 2274, 2275, 2276, 2277, 2278, 2279, 2280, 2281, 2282, 2283, 2284, 2285, 2286, 2287, 2288, 2289, 2290, 2291, 2292, 2293, 2294, 2295, 2296, 2297, 2298, 2299, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 2309, 2310, 2311, 2312, 2313, 2314, 2315, 2316, 2317, 2318, 2319, 2320, 2321, 2322, 2323, 2324, 2325, 2326, 2327, 2328, 2329, 2330, 2331, 2332, 2333, 2334, 2335, 2336, 2337, 2338, 2339, 2340, 2341, 2342, 2343, 2344, 2345, 2346, 2347, 2348, 2349, 2350, 2351, 2352, 2353, 2354, 2355, 2356, 2357, 2358, 2359, 2360, 2361, 2362, 2363, 2364, 2365, 2366, 2367, 2368, 2369, 2370, 2371, 2372, 2373, 2374, 2375, 2376, 2377, 2378, 2379, 2380, 2381, 2382, 2383, 2384, 2385, 2386, 2387, 2388, 2389, 2390, 2391, 2392, 2393, 2394, 2395, 2396, 2397, 2398, 2399, 2400, 2401, 2402, 2403, 2404, 2405, 2406, 2407, 2408, 2409, 2410, 2411, 2412, 2413, 2414, 2415, 2416, 2417, 2418, 2419, 2420, 2421, 2422, 2423, 2424, 2425, 2426, 2427, 2428, 2429, 2430, 2431, 2432, 2433, 2434, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2442, 2443, 2444, 2445, 2446, 2447, 2448, 2449, 2450, 2451, 2452, 2453, 2454, 2455, 2456, 2457, 2458, 2459, 2460, 2461, 2462, 2463, 2464, 2465, 2466, 2467, 2468, 2469, 2470, 2471, 2472, 2473, 2474, 2475, 2476, 2477, 2478, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2487, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2520, 2521, 2522, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2583, 2584, 2585, 2586, 2587, 2588, 2589, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2620, 2621, 2622, 2623, 2624, 2625, 2626, 2627, 2628, 2629, 2630, 2631, 2632, 2633, 2634, 2635, 2636, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2651, 2652, 2653, 2654, 2655, 2656, 2657, 2658, 2659, 2660, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2673, 2674, 2675, 2676, 2677, 2678, 2679, 26

# Interactions need a domain meaning

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## Tomato example

$x_1 x_2$  lets the effect of color depend on firmness.

## Interpretation

Deep color may support pack now only when the tomato is not unusually hard.

## Spurious interaction

A large coefficient can reflect chance correlation when data are scarce.

# A richer map can also fit noise

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## Training fit

A high degree polynomial can pass close to many training points.

## Between the points

The boundary may oscillate in regions with little evidence.

## Model selection

Use validation performance, regularization, and a degree chosen before looking at the test set.

quickly  
must be chosen  
on preprocessing

```
import numpy as np

X = np.array([[1., 2.],
              [3., 4.]])
x1, x2 = X[:, 0], X[:, 1]
Phi = np.column_stack([
    x1**2, x1*x2, x2**2, x1, x2, np.ones(len(X))
])
print(Phi)
print(Phi.shape)
```

# Interval classifier in code

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```
a, b = 3.0, 1.0
x = np.array([0.0, 1.5, 2.0, 3.5])
Phi = np.column_stack([x**2, x, np.ones(len(x))])
w = np.array([1.0, -(a + b), a * b])

scores = Phi @ w
labels = np.where(scores >= 0, 1, 2)
print(scores)
print(labels)
```

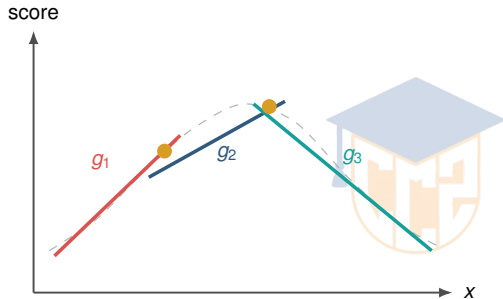
## Expected labels

Class 1, class 2, class 2, class 1.

Use several affine pieces, each responsible for a local part of the input space.

Straight segments join to approximate a curved or multi region decision surface.

The number and placement of pieces must be learned or chosen.



Local affine pieces approximate a curved score while keeping each piece simple.

# Several local lines fit a multimodal class

## Two tomato varieties

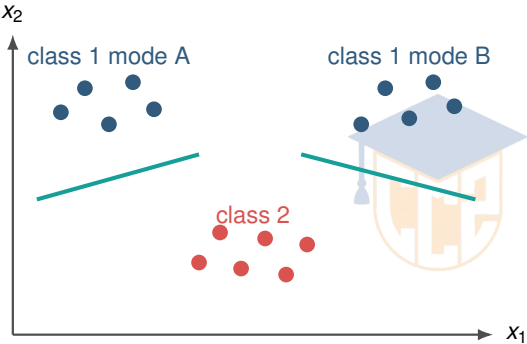
The pack now class forms two separated clusters because the varieties mature at different colors.

## One global line

It may cut through the hold cluster or miss one pack cluster.

## Local pieces

Each cluster can receive a nearby linear comparison.



Two local separators handle a class with two separated modes.

| Boundary family           | Strength                | Main limitation               |
|---------------------------|-------------------------|-------------------------------|
| One affine boundary       | Simple and stable       | Cannot bend                   |
| Feature mapped boundary   | Smooth global curvature | Feature count may grow        |
| Piecewise affine boundary | Local flexibility       | Pieces and joins need control |

Choose the smallest family that captures repeatable structure.



$$g_{c1}(x) = g_{c2}(x)$$

$$(w_{c1} - w_{c2})^\top x + (b_{c1} - b_{c2}) = 0$$

The active piece changes across this hyperplane.

Both pieces may belong to the same class.

# Nearest prototypes create affine comparisons

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## Choose the nearest prototype

$$\hat{j} = \arg \min_j \|x - \mu_j\|^2$$

$$-\|x - \mu_j\|^2 = -x^\top x + 2\mu_j^\top x - \|\mu_j\|^2$$



## Shared term

$-x^\top x$  cancels when prototypes are compared.

# Several prototypes bend the class boundary

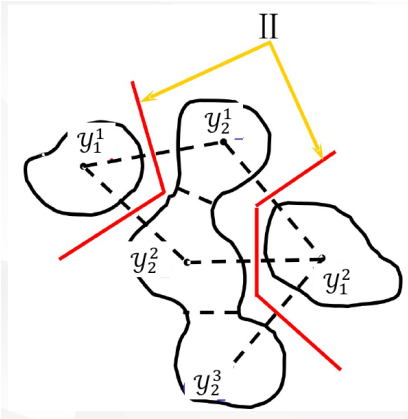
## Subclasses

Class  $c$  has prototypes  $\mu_{c1}, \dots, \mu_{cL_c}$ .

$$d_c(x) = \min_{\ell} \|x - \mu_{c\ell}\|^2, \quad \hat{y} = \arg \min_c d_c(x)$$

## Geometry

Prototype Voronoi cells meet along line segments. Class labels merge neighboring cells.



# A piecewise boundary can be built in stages

## Start

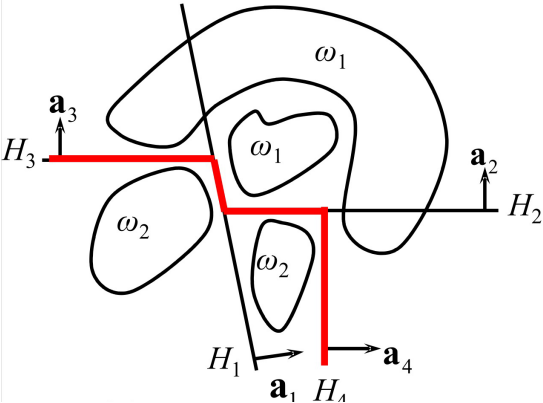
Find one difficult local comparison between opposite classes.

## Refine

Add another separating plane where errors remain.

## Stop

End when validation improvement no longer justifies another piece.



| Pieces         | Likely behavior                      |
|----------------|--------------------------------------|
| 1              | A single affine boundary             |
| A few          | Captures broad local structure       |
| Many           | Tracks fine variations and noise     |
| One per sample | Approaches nearest neighbor behavior |

## Selection

Choose the piece count with validation evidence, not training error alone.



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Merge prototypes, require minimum support, or penalize model complexity.

## Encounter regions contain the hard comparisons

## Encounter region

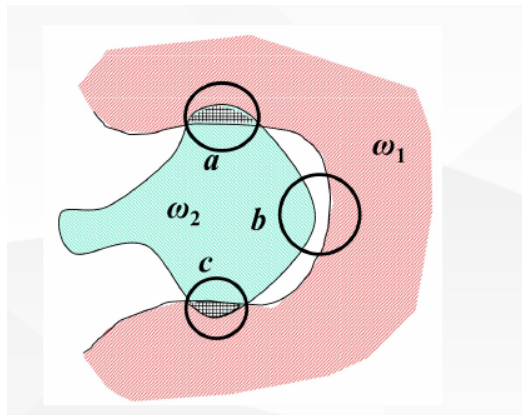
A local area where samples or prototypes from different classes are close.

## Training focus

Use nearby opposite class evidence to design the local separator.

## Coverage

Easy interior points still matter for defining where the local rule is allowed to act.

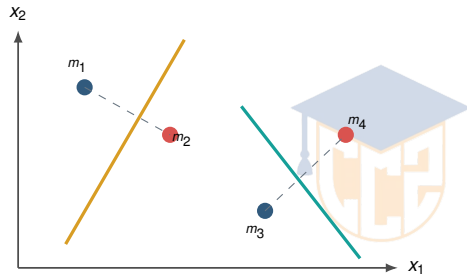


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A prototype from each class is the other's nearest opposite class prototype.

$$\left[ x - \frac{\mu_a + \mu_b}{2} \right]^\top (\mu_a - \mu_b) = 0$$

Use the perpendicular bisector, then refine with nearby samples.



A perpendicular bisector gives an initial local plane for each opposite-class prototype pair.

The distance matrix has three samples by four prototypes.

ear discriminant

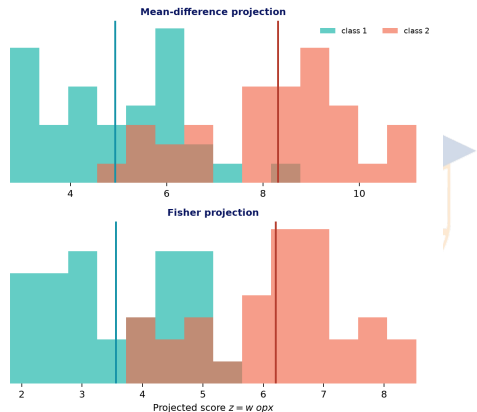
## Goal

Project many features onto one axis while preserving class separation.

$$z = w^T x$$

## Fisher criterion

Projected class centers should be far apart and each class should remain compact.



# How can we preserve or destroy secrets?

Same points

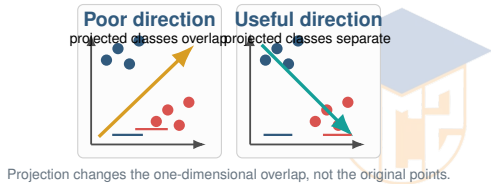
Only the direction of projection changes.

Poor direction

Different classes land on overlapping positions.

Useful direction

Class centers separate and within class spread stays small.

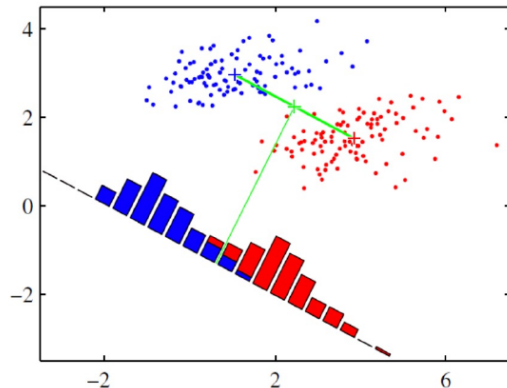


Projection changes the one-dimensional overlap, not the original points.

The projected histograms overlap strongly.

The axis preserved a direction with large within class variation.

**Lesson.** Separation in the original plane does not survive every one dimensional view.

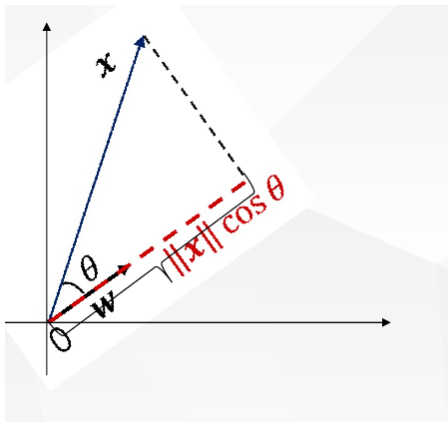




If  $\|w\| = 1$ , then  $z = w^\top x = \|x\| \cos \theta$ .

Replacing  $w$  by  $cw$  rescales every projected value by  $c$ .

**Geometry.** The perpendicular foot on the axis determines the scalar coordinate.



Original mean

$$m_k = \frac{1}{N_k} \sum_{i: y_i=k} x_i$$

## Projected mean

$$\tilde{m}_k = \frac{1}{N_k} \sum_{i: y_i=k} w^\top x_i = w^\top m_k$$

## Consequence

Projected center difference is  $w^\top (m_1 - m_2)$ .

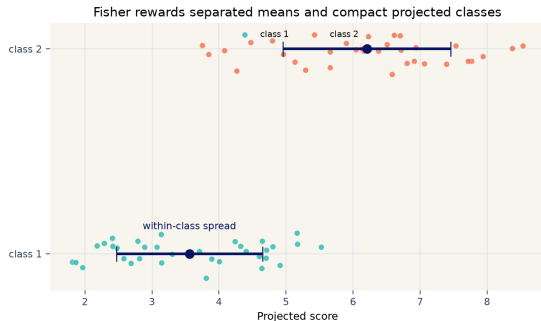
## Class scatter

$$S_k = \sum_{i: y_i = k} (x_i - m_k)(x_i - m_k)^T$$

## Projected scatter

$$\tilde{\mathbf{S}}_k^2 = \sum_{i: y_i = k} (\mathbf{w}^\top \mathbf{x}_i - \mathbf{w}^\top \mathbf{m}_k)^2 = \mathbf{w}^\top \mathbf{S}_k \mathbf{w}$$

Total within scatter

 $w^\top S_W w$  where  $S_W = S_1 + S_2$ .



$$(\tilde{m}_1 - \tilde{m}_2)^2 = [w^\top (m_1 - m_2)]^2$$

$$S_B = (m_1 - m_2)(m_1 - m_2)^\top$$

$$[w^\top (m_1 - m_2)]^2 = w^\top S_B w$$

$$J(w) = \frac{w^\top S_B w}{w^\top S_W w} = \frac{[w^\top (m_1 - m_2)]^2}{w^\top S_W w}$$

# The Fisher ratio ignores scale

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## Rescale the direction

Let  $w' = cw$  for any nonzero  $c$ .

$$J(cw) = \frac{c^2 w^\top S_B w}{c^2 w^\top S_W w} = J(w)$$



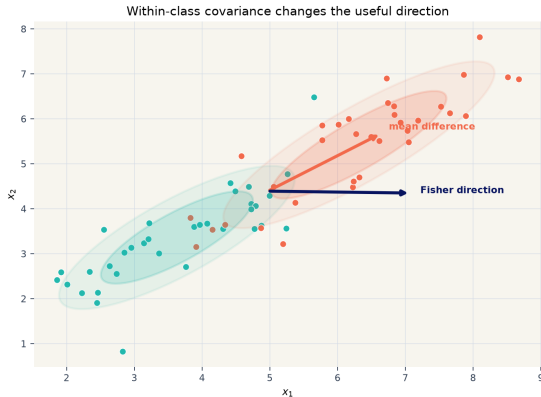
## Practical normalization

Use  $w/\|w\|$  for display, or fix  $w^\top S_W w = 1$  in a derivation.

$$W = m_1 - m_2$$

It ignores directions where each class has large variance.

The center line may align with a noisy long axis and create overlapping projections.



Maximize  $w^\top S_B w$  subject to  $w^\top S_W w = 1$ .

**Two class simplification.**  $S_B w = (m_1 - m_2)(m_1 - m_2)^\top w$  is a scalar multiple of  $m_1 - m_2$ .



$$w \propto S_W^{-1}(m_1 - m_2)$$

A modest mean difference can matter when measurements are consistent.

- 1 Compute  $m_1$  and  $m_2$  from their own classes.
- 2 Center every sample using its class mean.
- 3 Accumulate outer products inside each class.
- 4 Add the two matrices to obtain  $S_W$ .

### Common mistake

Centering all samples by the global mean mixes between class separation into the within class scatter.

$$w = S_W^{-1}(m_1 - m_2)$$

Solve  $S_W w = m_1 - m_2$ .

```
Sw = np.array([[4., 1.], [1., 2.]])
mean1 = np.array([1., 1.])
mean2 = np.array([3., 2.])
w = np.linalg.solve(Sw, mean1 - mean2)
```

Explicit matrix inversion is slower and usually less numerically stable.

# A two dimensional Fisher example

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## Class means

$$m_1 = (2, 0)^T \text{ and } m_2 = (2, 2)^T$$

## Within class scatters

$$S_1 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix} \text{ and } S_2 = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$$

$$S_W = 2I, \quad w = S_W^{-1}(m_1 - m_2) = (0, -1)^T$$

# Projected scores in the numeric example

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| Point    | $z = w^T x$ | Side of midpoint |
|----------|-------------|------------------|
| (2, 0)   | 0           | Class 1 center   |
| (2, 2)   | -2          | Class 2 center   |
| (4, 0.4) | -0.4        | Class 1 side     |
| (1, 1.6) | -1.6        | Class 2 side     |



## Midpoint threshold

$$t = (0 - 2)/2 = -1.$$

Fisher selects  $w$  to separate one dimensional class distributions.

Choose a threshold  $t$  and compare  $z = w^\top x$  with  $t$ .

The midpoint  $(w^\top m_1 + w^\top m_2)/2$  is a common choice.

The Fisher criterion alone does not encode priors or unequal error costs.

# Class priors move the projected threshold

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## Equal projected variance

If  $z \mid k \sim \mathcal{N}(\mu_k, \sigma^2)$ , Bayes gives

$$t = \frac{\mu_1 + \mu_2}{2} - \frac{\sigma^2}{\mu_1 - \mu_2} \log \frac{\pi_1}{\pi_2}$$



## Interpretation

A more common class receives a larger region along the projection axis.



$$x \mid y = k \sim \mathcal{N}(\mu_k, \Sigma)$$

$$W_{\text{Gaussian}} = \Sigma^{-1}(\mu_1 - \mu_2)$$

Fisher uses  $S_W^{-1}(m_1 - m_2)$ , which has the same form.

Priors and losses still determine the intercept of the Gaussian decision rule.

Thousands of pixels with only dozens of labeled images cannot produce a full rank within class scatter.

A direct solve fails or becomes extremely sensitive to small perturbations.

# Regularization stabilizes the Fisher direction

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## Regularized scatter

$$S_W^{(\lambda)} = S_W + \lambda I, \quad \lambda > 0$$

## Direction

$$w_\lambda = (S_W + \lambda I)^{-1}(m_1 - m_2)$$

## Effect

Small variance directions no longer receive arbitrarily large inverse weights.

## Selection

Choose lambda with validation data.

The normalized vector points almost straight down.

Model checks 数字工程研究中心

- 1 Compare training and validation separation on the projected axis.
- 2 Inspect projected distributions rather than only their means.
- 3 Report the threshold, priors, and loss assumptions separately.
- 4 Stress test the direction after small data and scaling changes.

## A high criterion is not enough

A few outliers or a shifted deployment population can still produce poor decisions.

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| Design           | What is learned                           | Best first question                       |
|------------------|-------------------------------------------|-------------------------------------------|
| Linear           | one score $w^\top x + w_0$                | Can one straight boundary work?           |
| Feature map      | a linear score in $\phi(x)$               | Which transformation exposes the pattern? |
| Piecewise linear | several local linear scores               | Does the boundary need several faces?     |
| Fisher           | a projection with low within class spread | Which direction separates compactly?      |

Begin with the simplest geometry that can express the decision.

Size separates most samples, but two growing regions create a bent boundary in color and firmness.

Choose the threshold from the cost of sending a wrong variety to packaging.

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We designed decision functions and understood their geometry.

The perceptron will learn a separating direction from mistakes; decision trees will partition the space by questions.

A learning rule is not only an optimizer. It also determines which boundaries are easy to discover.

## Exit ticket先进计算与数字工程研究中心

- 1 State the difference between a generative and a discriminative classifier in one sentence.
- 2 Explain the sign and magnitude of a linear discriminant score.
- 3 Give one reason to use a feature map and one reason to use linear pieces.
- 4 Explain the numerator and denominator of the Fisher criterion.
- 5 For XOR, choose a model family and justify the choice.

## A strong answer

Names the geometry, the information used, and one failure mode rather than listing formulas.